

How Much Water Does It Take to Get a Shirt Wet?

This really isn't a good question to ask in the context of washing clothing. Why are we adding water to clothing?

Water cleans clothing through a variety of means. Properties of water such as its mass, viscosity, and surface tension are factors which can be utilized to mechanically carry contaminants away from clothing. Water can carry and disperse cleaning agents such as soaps, detergents, and disinfectants, which can help to clean or disinfect clothing by evenly diffusing across the clothing item. Water also dissolves some forms of contamination. In each of these stated cases, water is added to a clothing item and made to mix with the contaminants. Contaminants present in this mixture can only be removed by being carried away with the water when removed.

For a clothing item, only so much water can be removed mechanically for cleaning. This is why you have a washing machine and dryer at home. Because of this, for whatever wash process is chosen, at a minimum, there is an amount of water that needs to be added to a clothing item before any cleaning can be accomplished. I called this volume of water dead volume.

Clothing Volume Characterization

A clothing item itself has volumes that can be characterized differently. For example, you can pack a shirt in a box. Maybe when loosely packed it takes up all the space in the box, and you can easily close the lid. I called this heap volume and made for a good metric of wash chamber size that allows a user to easily insert a clothing item into a machine.

Of course, a shirt can be made to fit into a smaller box if you stuff it in more, this could be another form of clothing volume, but how should you push down on the clothing item. You can only compress it so much, and maybe past a certain point you'll damage the fabric. The minimum volume for the maximum force you'd apply to a clothing item is the clothing's compressed volume.

Even if you compress clothing as much as you can, there is still some space between all the clothing fibers. If you could compress the clothing item so there were no pockets of air or water, this would be the true volume of the clothing item. You can also find this by submerging the clothing in water and observing the displacement. This is the true volume of the clothing item.

You can determine the dead volume by observing how much water you recover from a drying process (compression, wringing, flinging, etc.) and comparing it to the amount of water added prior to the process. You can also infer the dead volume if you know the true volume of a clothing item and the compressed volume. Different methods of measurement may be used or be better suited depending on the wash process being tested or implemented. Regardless of the wash process, if water is being used, it is likely that a process that minimizes dead volume will utilize water most effectively.

Types of Contaminants & Washing Considerations

Clothing can become dirty in a variety of ways, but regardless of how clothing is soiled, it is because contaminants are trapped within the fabric of the clothing item. For the purposes of washing, how can these contaminants be categorized?

State of matter: Solids and Liquids

Solubility: Water Soluble, and non-Soluble

Other Factors: Bacteria (Not in scope of project as this is fixed with soap/disinfectants)

Think of what sorts of contaminants you might find in a space-craft environment. In what way will a clothing item become dirty? On the ISS as an example, it might be expected that most of the contaminants found on a clothing item would be water soluble salts from sweat. The scope of this design is not for astronauts going to the moon, where regolith, a non-soluble dust is found.

For the types of contaminants you expect to see, what wash process may be best suited to clean clothing? If you expect non-soluble solids, you will need readily open pathways for water to carry these particles away from the clothing item. If you do not expect to see this, that may not be of concern.